CSE341: Microprocessor

Quiz - 2 **Section:** **Marks: 15** **Duration: 25 minutes**

Name: **ID:**

Question-1:

20-bit Physical Address	2427F	2427E	2427D	2427C	2427B	2427A	24279
8-bit Data (in HEX)	83	BA	42	7A	83	DA	FF

Consider the following instructions: **JMP [BX+1000h]**
The physical address after executing the above JMP instruction is **3427Ah.** For this question, use **DS** = 2200h, **SI**=1246h, **SS**=3000h, **CS**=3000h, **DI** = 1000h, and **use** the portion of memory given above.

a. **State** the addressing mode of the given instruction. [2 marks]
b. **Calculate** the content of the BX register. [5 marks]

a. Addressing Program Codes (relative register with displacement).

b. CS x 10 + IP = 3427Ah
3000 x 10 + IP = 3427Ah
IP = 42**7A**h
The Low byte of IP's (7A) address = 2427C [From Memory Table]

DS x 10 + BX + 1000 = 2427C
2200 x 10 + BX + 1000 = 2427C
BX = 2427C - 1000 - 22000
BX = 127C

Question-2: Suppose you are given the instructions: **MOV DX, [BX+DI+4FFh]** and the opcode for **MOV = 100010**

a. Deduce the full machine code for given instruction. Your answer should be in hex. [5 marks]
b. Estimate the machine code length (in bytes) for the given instruction. [1 mark]

a. MOV DX, [BX+DI+4FFh] → MOV DX, [BX+DI+04FFh]

→ 100010 1 1 10 010 001 1111 1111 0000 0100 → 8B91FF04h

b. Mod = 10 (16 bit displacement). So, the machine code length = 4 bytes.

Question-3: Consider two instructions: **i. MOV AX, 1234h , ii. MOV BX, 5678h**. Perform ADD AX, BX and determine the values of OF, PF, AF, SF, ZF, and CF, explaining the reasoning for each. [4 marks]

- 1234h → 0001 0010 0011 0100
- 5678h → 0101 0110 0111 1000
  → 0110 1000 1010 1100

- CF = 0 [No carry out occurs from the 16-bit addition]
- OF = 0 [No carry in and no carry out occurs from the 16-bit]
- PF = 1 [Even number of 1s in Low byte]
- AF = 0 [No carry out occur from lower nibble]
- SF = 0 [MSB is 0]
- ZF = 0 [Result is not zero]